BAAO COMMUNITY COLLEGE
SAN JUAN BAAO CAMARINES SUR

FINAL PROJECT IN CC106-APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES

GROUP 4: SYNTACTIC SYNDICATE

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PROPOSED APPLICATION/SYSTEM

BARANGAY YOUTH INFORMATION & SERVICE PORTAL

I. SYSTEM OVERVIEW

The Barangay Youth Information & Service Portal (BYISP) is a web-based database management system designed to digitize, centralize, and automate youth demographic profiling and document service request workflows for barangay administration.

The system developed using PHP, MySQL, and Laragon, Barangay Youth Information & Service Portal (BYISP) addresses the operational bottlenecks of manual paper logbooks by introducing a direct-access management portal. The system features dynamic walk-in resident profile registration, multi-criteria filtering, 3rd Normal Form (3NF) relational database integrity, and automated CSV data exports for administrative reporting and auditing.

II. PROJECT BACKGROUND & STATEMENT OF THE PROBLEM

A. Current Context

Barangay youth administration relies heavily on physical logbooks and paper forms to register residents, record sports/skolarship applications, and track barangay clearings or service requests.

B. Problem Statement

1. **Data Redundancy & Fragmentation:** Resident records are scattered across physical paper files, leading to lost forms and duplicate entries.
2. **Delayed Processing:** Administrative officers face slow search and retrieval times when verifying resident eligibility for service requests.
3. **Lack of Analytics & Reporting:** Generating summary reports for youth demographics and service distribution requires manual, time-consuming tallies.

III. PROJECT OBJECTIVES

A. General Objective

To design, develop, and deploy a functional, responsive web portal that automates youth profiling and document service request workflows for local barangay governance.

B. Specific Objectives

1. Automated Profiling: Implement a relational database schema (3rd Normal Form) that links resident demographic data directly to service requests.
2. Dynamic Request Processing: Provide an interface for entering and processing document requests with auto-handling for existing and walk-in residents.
3. Search & Pagination Filtering: Enable instant record filtering by resident name, purok/zone, category, or request status (Pending, Approved, Rejected) with paginated navigation.
4. Data Portability: Enable one-click CSV exporting for offline reporting, administrative auditing, and record keeping.

IV. SYSTEM SCOPE AND LIMITATIONS

A. System Scope

Direct-Access Dashboard: Streamlined administrator interface bypassing session log-in barriers for rapid record management.

Service Request Management (CRUD): Full Create, Read, Update, and Delete operations for service requests.

Auto-Profile Generation: Intelligent form logic that detects existing residents or automatically generates user and profile records for new walk-ins.

Export Engine: Dynamic CSV file generator outputting clean tabular data filtered by search criteria

B. System Limitations

System operates on a local server environment (Laragon / Localhost) using PHP and MySQL.

Mobile SMS notifications and online payment gateways are excluded from the current prototype scope.

V. TECHNICAL ARCHITECTURE & SYSTEM SPECIFICATIONS

A. Software & Development Stack

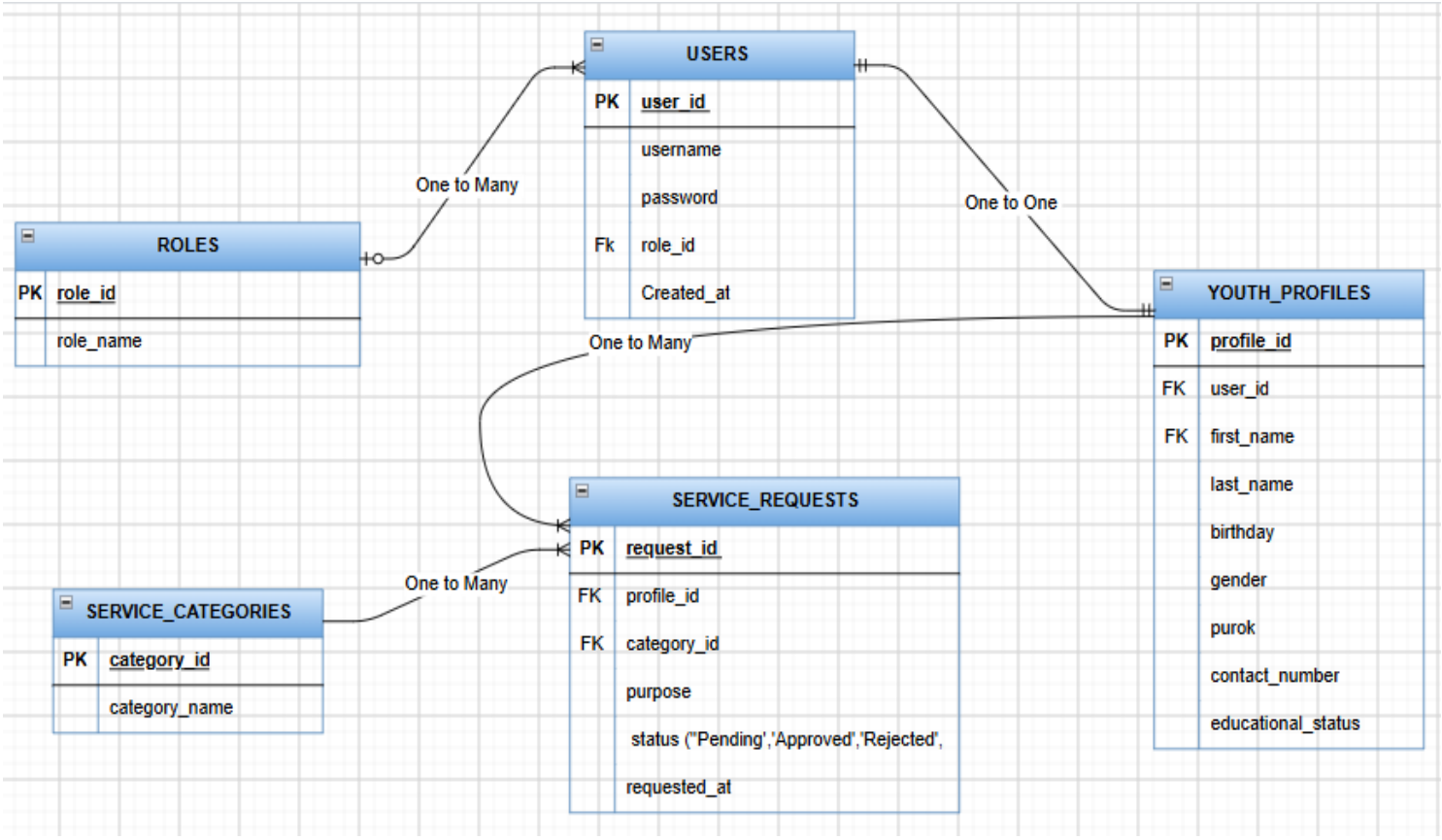
Frontend: HTML5, CSS3, Bootstrap 5.3 (Responsive Framework

Backend: PHP (Procedural with Prepared Statements / mysqli)

Database Engine: MySQL Server (Managed via HeidiSQL)

Development Environment: Laragon Local Web Server Stack

B. Database Schema (Normalized to 3NF)



Entity Cardinality between related tables:

1. **ROLE to USERS - One-to-One (1:1)**
   - Each user account is linked to strictly one resident youth profile. This is enforced in database schema where user_id in youth_profile has a unique constraint.

2. **USERS to YOUTH PROFILE - One-to-One (1:1)**
   - Each user account I linked to strictly one resident youth profile,and each youth profile corresponds to exactly one user account login, this enforced in your database schema where user_id in youth_profile has a unique constraints.

3. **YOUTH_PROFILES TO SERVICE_REQUESTS - ONE-TO-MANY (1:N)**
   - A single resident profile (youth_profiles) can submit and own multiple service or document requests over time (service_requests), but each individual service request is owned by exactly one resident profile.

4. **SERVICE_CATEGORIES TO SERVICE_REQUESTS - ONE-TO-MANY (1:N)**
   - A single service category (e.g., Educational Assistance, Sports & Recreation) can be selected across many different service requests, but each individual service request falls under exactly one category.

3rd Normal Form (3NF) Justification

1NF: Every column contains atomic (indivisible) values. Tables contain primary keys **(role_id, user_id, profile_id, category_id, request_id).**

2NF: The schema is in 1NF and all non-key attributes are fully dependent on their respective primary key.

3NF: No transitive dependencies exist. Attributes like **category_name** are extracted into **service_categories** rather than repeating inside **service_requests.** purok and demographic info depend strictly on **profile_id.**

VI.DATA DICTIONARY

Table Name	Column Name	Data Type	Constraint	Description
ROLES	role_id	INT	PK, AUTO_INCREMENT	Unique ID for user role
	role_name	VARCHAR (50)	NOT NULL, UNIQUE	Role name (Admin, Youth_User)

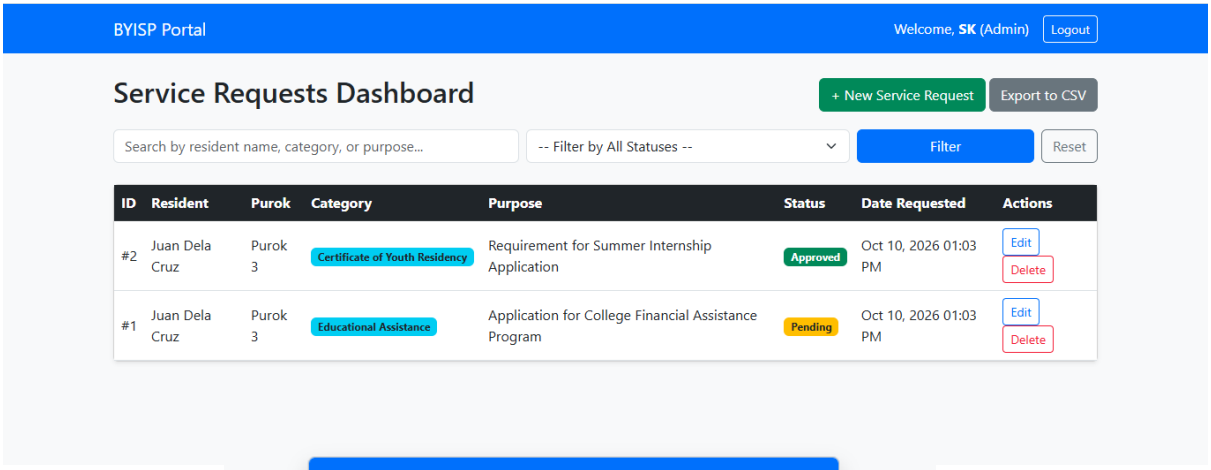
Table Name	Column Name	Data Type	Constraint	Description
USERS	user_id	INT	PK, AUTO_INCREMENT	Unique user account ID
	username	VARCHAR (50)	NOT NULL, UNIQUE	User login name
	password	VARCHAR (255)	NOT NULL	BCRYPT hashed password hash
	role_id	INT	FK -> roles. Role_id	User role assignment
	created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration timestamp

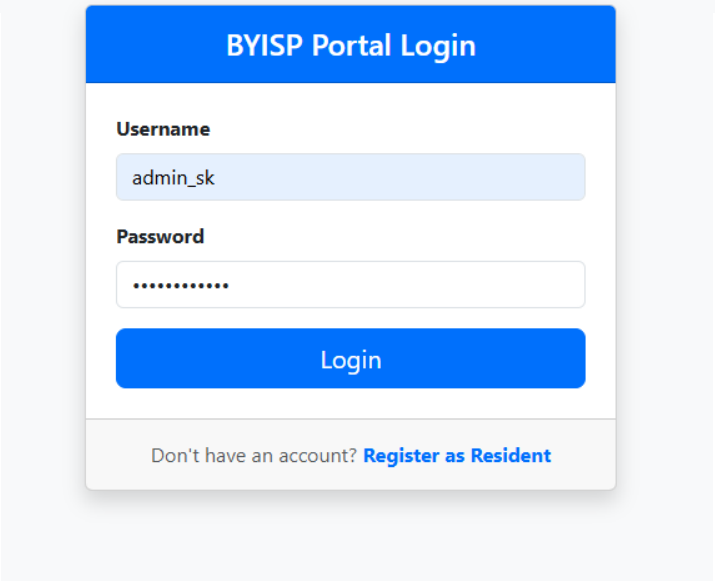
Table Name	Column Name	Data Type	Constraint	Description
service_categories	category_id	INT	PK, AUTO_INCREMENT	Service category ID
	category_name	VARCHAR (100)	NOT NULL	Title of service
	description	TEXT	NULLABLE	Details about the service

Table Name	Column Name	Data Type	Constraint	Description
YOUTH_PROFILES	profile_id	INT	PK, AUTO_INCREMENT	Primary key for youth profile
	user_id	INT	FK -> users.user_id, UNIQUE	Linked system account
	first_name	VARCHAR (50)	NOT NULL	Resident given name
	last_name	VARCHAR (50)	NOT NULL	Resident surname
	birthdate	DATE	NOT NULL	Birthdate
	gender	ENUM	'Male', 'Female', 'Other'	Gender identification
	purok	VARCHAR (50)	NOT NULL	Barangay sub-zone/purok
	contact_number	VARCHAR (15)	NOT NULL	Mobile number
	educational_status	VARCHAR (50)	NOT NULL	e.g., Enrolled, Out of School Youth

Table Name	Column Name	Data Type	Constraint	Description
service_requests	request_id	INT	PK, AUTO_INCREMENT	Request reference ID
	profile_id	INT	FK -> youth_profiles.profile_id	Submitting resident
	category_id	INT	FK -> service_categories.category_id	Category of requested service
	purpose	TEXT	NOT NULL	Detailed reason for request
	status	ENUM	'Pending', 'Approved', 'Rejected'	Processing state
	requested_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Application date

DOCUMENTATION OF PROPOSED WED APPLICATION/SYSTEM





Conclusion

In conclusion, the Barangay Youth Information & Service Portal successfully transitions local barangay governance from error-prone manual logs to a secure, organized, and data-driven digital workflow. as a future development, we recommend deploying the system to a live cloud server, integrating SMS notification alerts for residents, and adding secure online account portals for youth residents themselves.